

DDTA - Scheda di lettura compilata

SRC-0040 - BA0-R systems-modeling prior art

REVIEW-ONLY: da approvare prima del caricamento nel repository.

Identita bibliografica e accesso	
Source ID	SRC-0040
Titolo	OMG Systems Modeling Language (SysML) Version 2.0 - Language Specification
Autori/ente	Object Management Group (OMG)
Anno	2025
Identificatore	SysML 2.0; formal adoption September 2025; current normative Language PDF OMG file id formal/26-03-02
Accesso	public_normative_specification
Verification state	verified_full_text_public

1. Research problem and contribution

Defines the semantics and notation of SysML v2 as a general-purpose systems modeling language built on KerML. It provides formal concepts for items, parts, ports, connections, interfaces, flows, actions, states, requirements, cases, views/viewpoints and analysis.

2. Starting artifacts and generated representations

A part is a structural item; ports are connection points; connections relate usages and may have their own properties; interfaces are specialized connections whose ends are ports; flows model transfer of payloads between source and target; actions model behavior over time; states describe modes of occurrence behavior. Views expose, filter and render portions of a model.

3. Traceability, change, automation and human role

The language supports explicit relationships such as satisfy, allocation, exposure/import, view filtering and rendering. A rendered view artifact can preserve correspondence back to model elements. Analysis cases can specify what analysis is to be performed while an external solver controls execution details.

4. Findings, limitations and relationship with DDTA

Actor is not a universal root: an actor parameter is a role played by an external entity and is a PartUsage. Asset is likewise not the generic system object; SysML uses Item/Part semantics. Connection, interface, flow and interaction are distinct. The AnalysisCase construct substantially narrows any DDTA novelty claim around a generic analysis envelope. DDTA must differentiate itself through governed documentation grounding, method overlays, reviewed findings, accepted document change and re-analysis/provenance rather than the existence of an analysis container.

Evidenza utile per BA0/BA1

Pag.	Ruolo	Sezione	Interpretazione DDTA
89	challenge	7.11 Parts Overview	A generic structural element is more neutral than Actor as a root.
92	foundation	7.12 Ports Overview	Endpoint/interface semantics have an independent responsibility.
95	foundation	7.13 Connections Overview	A channel-like concept may be a specialization/role of a more generic first-class connection.
105	foundation	7.14 Interfaces Overview	Interface should not be equated with flow or payload.

Pag.	Ruolo	Sezione	Interpretazione DDTA
111	foundation	7.15 Flows Overview	Payload, flow and endpoints are semantically separable.
144	foundation	7.18 States	State/mode is explicit behavior semantics, not merely an annotation.
161	challenge	7.21 Requirements	Actor is context-dependent role semantics in SysML, weakening Actor as a generic BAE root.
169	challenge	7.23 Analysis Cases	Generic analysis specification is established prior art.
171	challenge	7.23.3 Trade-Off Analyses	Analysis and trade-study semantics are not DDTA-specific contributions.
180	foundation	7.26 Views and Viewpoints	Strong support for canonical-model-to-view projection.

Bibliography mining / follow-up

- OMG KerML 1.0
- SysML v2 Analysis Domain Library
- SysML v2 Requirement Derivation Domain Library

Governance note

Questa scheda è un ausilio di review. Le future citazioni di tesi devono risolversi alla fonte registrata e a una posizione esatta nell'excerpt ledger approvato. Nessun concetto esterno diventa automaticamente un BAE DDTA.